03 The Resource Room

Where carrying capacity actually comes from

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 33 min	**+ 30 min**	The Crowding Room (02), where K was simply given.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Resource tab

A resource is supplied at rate S, decays at rate l and is eaten by consumers, whose births depend on how much they eat. Nothing in the equations is a carrying capacity — and yet one appears. This guide asks where it comes from and what sets it.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain how a carrying capacity emerges from a supplied and consumed resource.
- Define R^* as the resource level at which a consumer breaks even, and read it from the room.
- Predict that richer supply raises the number of consumers but not R^*.
- Explain why carrying capacity is a property of a species *in an environment*, not of the species alone.

The extension adds

- Calculate $R^* = m/(ea)$ and $N^* = (S - lR)/(aR)$ by hand and check them.
- Predict when the approach to equilibrium is a damped spiral and when it is smooth.
- State the condition $S/l > R^*$ for the consumer to persist.

SUGGESTED ROUTES

- **35-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Problem class:** steps 5 and 7 on paper first, then checked in the room.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 The ceiling nobody typed in

10 min

SET UP

The defaults: S = 15, l = 0.02, R_0 = 70, a = 0.008, e = 0.80, m = 0.32, N_0 = 5, time span 120.

PREDICT FIRST

As consumers build up, what will happen to the resource? Will the consumers keep growing, crash, or settle?

RUN

Press **Run**.

LOOK AT

the dashed R^* line on *Resource over time*, the dashed N^* line on *Consumers over time*, and the readouts in *The emergent ceiling*.

EXPLAIN

Every time the consumer curve turns over, where is the resource? Why?

2 Double the supply

8 min

SET UP

S = **30**.

PREDICT FIRST

What will happen to *R? To N*?

RUN

Run.

EXPLAIN

Why does enriching the environment change how many consumers it holds but not the resource level they leave behind?

3 The environment shifts

8 min

SET UP

S back to 15. Open *More parameters*, **Environment shift**: On, multiplier 0.50.

PREDICT FIRST

Halfway through, supply halves. What will happen to the consumers, and to the resource, after the shift?

RUN

Run and use the History Scrubber around the halfway point.

EXPLAIN

The consumers have no new traits and get no warning. What do they “track”?

4 A tougher consumer

7 min

SET UP

Shift off. m = **0.16** (half the default mortality).

PREDICT FIRST

What will R^* be now? Will there be more consumers or fewer?

RUN

Run.

EXPLAIN

What does a lower R^* mean in terms of what a consumer can survive on? (Keep this in mind for the Coexistence Room.)

EXTENSION · MSC

5 By hand, then by machine

12 min

SET UP

Defaults, but e = **0.40**.

PREDICT FIRST

Calculate $R^* = m/(ea)$ and $N^* = (S - lR)/(aR)$.

RUN

Run and compare with *The emergent ceiling*.

EXPLAIN

Which parameters appear in R, *and which only in* N? What does that tell you about which parts of the system each one is controlled by?

6 Spiral or slide?

10 min

SET UP

Defaults; then S = **40** and resource loss l = **0.40**, so that the resource now decays faster than the consumers die ($l > m$).

PREDICT FIRST

The room's methods say a spiral needs $m > l$ and $S^2 < 4mR(S - lR)$. Which setting should spiral into equilibrium?

RUN

Run both and compare the *Resource–consumer phase plane*.

EXPLAIN

Why does a quickly decaying resource damp out the oscillation?

7 A supply too poor to live on

8 min

SET UP

S = **1.0**, l = **0.05**, others at defaults.

PREDICT FIRST

Compare S/l with R^*. Can the consumer persist?

RUN

Run.

EXPLAIN

State the persistence condition in words. Why is it the resource the environment can maintain *without* consumers that matters?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc A lake receives twice as much phosphorus after farming intensifies upstream. According to this room, what happens to the algae-grazing zooplankton, and to the phosphorus level?
2. BSc Why is it misleading to say "the carrying capacity of a cow is 1 hectare"?
3. BSc Which consumer is the better competitor: one that eats a lot, or one with a low R^*? Explain.
4. MSc The resource here is abiotic, like a nutrient in a chemostat. What changes if the resource is itself a population that grows?
5. MSc The functional response is linear. What would a saturating response do to R^* and to the stability of the equilibrium?
6. MSc Explain how this model relates to the logistic: under what conditions does the consumer's dynamics look logistic?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** The resource crashes as consumers build up, then both settle in damped oscillations: $R^* = 50$ and $N^* = 35$. Each turn of the consumer curve coincides with the resource crossing R^*.
- **Step 2.** R^* stays at 50; N^* rises to about 72.5. Supply does not appear in R^*.
- **Step 3.** After the shift the consumers fall to about 16 while the resource returns to 50: the consumers track the supply, the resource is held at R^*.
- **Step 4.** R^* halves to 25; N^* rises to about 73. A low R^* means surviving on less.
- **Step 5.** $R^* = 0.32/(0.4 \times 0.008) = 100$; $N^* = (15 - 2)/(0.008 \times 100) \approx 16$. R^* depends only on the consumer; N^* also on the supply.
- **Step 6.** The default spirals in ($m > l$, and $225 < 896$). With $l = 0.4 > m$ there is no spiral: the consumers settle at $N^* = 50$ without circling. (Supply was raised so that S/l stays above R^*.)
- **Step 7.** $S/l = 20 < R^* = 50$, so the consumer cannot persist and dies out.